

First Homework report

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October 10, 2025

In this report, I will discuss two files: **statistics.ipynb** and **visualization.ipynb**. Both files relate to the **First Mandatory Homework** and cover **wrecked data** statistical analysis and corresponding visualizations.

1 Visual analysis

Here I trided my best to visualize and clean [1]. Below you can see what it contains:

	Animal type	Country	Weight kg	Body Length cm	Gender
0	–	–	–	–	–
1	–	–	–	–	–
2	European bison	Poland	930.000	335.0	male
3	European bison	Poland	909.000	311.0	not determined
4	European bison™	Poland	581.000	277.0	female
⋮	⋮	⋮	⋮	⋮	⋮
1006	red squirrel	Poland	0.346	20.0	female
1007	hedgehog	Germany	1.000	23.0	female
1008	hedgehog	Germany	0.500	17.0	female
1009	red squirrel	Poland	0.346	20.0	female
1010	hedgehog	Hungary	0.900	16.0	female

Table 1: Animal observation **wrecked data** part 1

	Animal code	Latitude	Longitude	Animal name	Observation date	Data compiled by
0	–	–	–	–	03.01.2024	James Johnson
1	–	–	–	–	03.02.2024	James Johnson
2	–	52.828845	23.820144	Szefu	01.03.2024	Anne Anthony
3	–	52.830509	23.826849	–	01.03.2024	Anne Anthony
4	–	52.834109	23.807093	–	01.03.2024	Anne Anthony
⋮	⋮	⋮	⋮	⋮	⋮	⋮
1006	–	52.212001	21.033187	Lola	7 May 2024	Anne Anthony
1007	–	49.561356	11.105334	–	7 May 2024	Bob Bobson
1008	–	49.561569	11.087046	–	7 May 2024	Bob Bobson
1009	–	52.212001	21.033187	Lola	7 May 2024	Anne Anthony
1010	–	47.509860	18.939943	–	8 May 2024	Anne Anthony

Table 2: Animal observation **wrecked data** part 2

Yep. It pretty cursed. I cleaned it:

1. Dropped rudimet columns.
2. Dropped lines with NaNs.

3. Stardartized objects manually renaming whem.

4. Multy negative values by -1 .

After this I visualized everithing.

1.1 Count plots

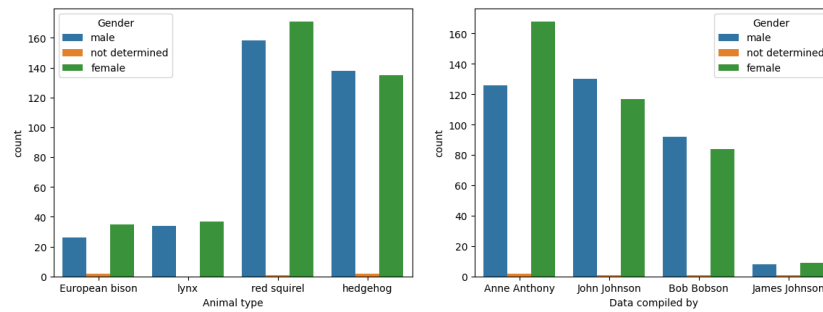


Figure 1: Count plots of animal type and of data contributor by gender.

On 1 gender distribution is normal but animal type is not as well as data contributing. Worth to take a closer look at James data.

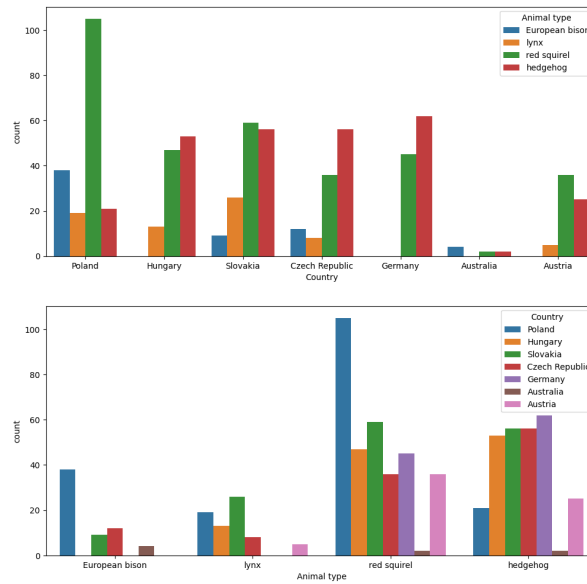


Figure 2: Count plot of animal type by country and vise versa.

On 2 plots I note nothing except that European bison lives in Australia.

1.2 Line plots

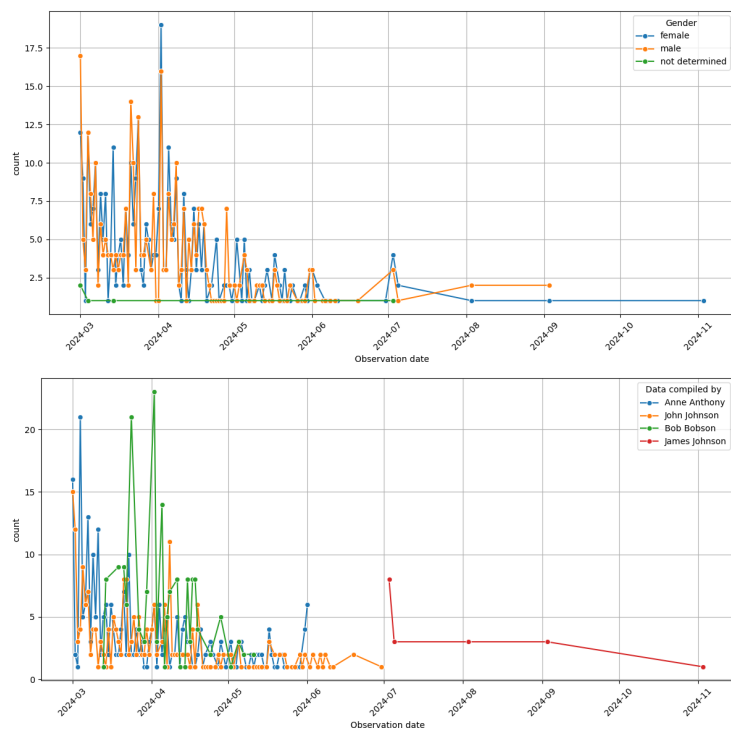


Figure 3: Line plots of observation date by gender and by data contributor

Rivising 3:

1. All data impliments in 2024.
2. James somewhat standing out even more.

1.3 Hist and box plots

2 Statistical analysis

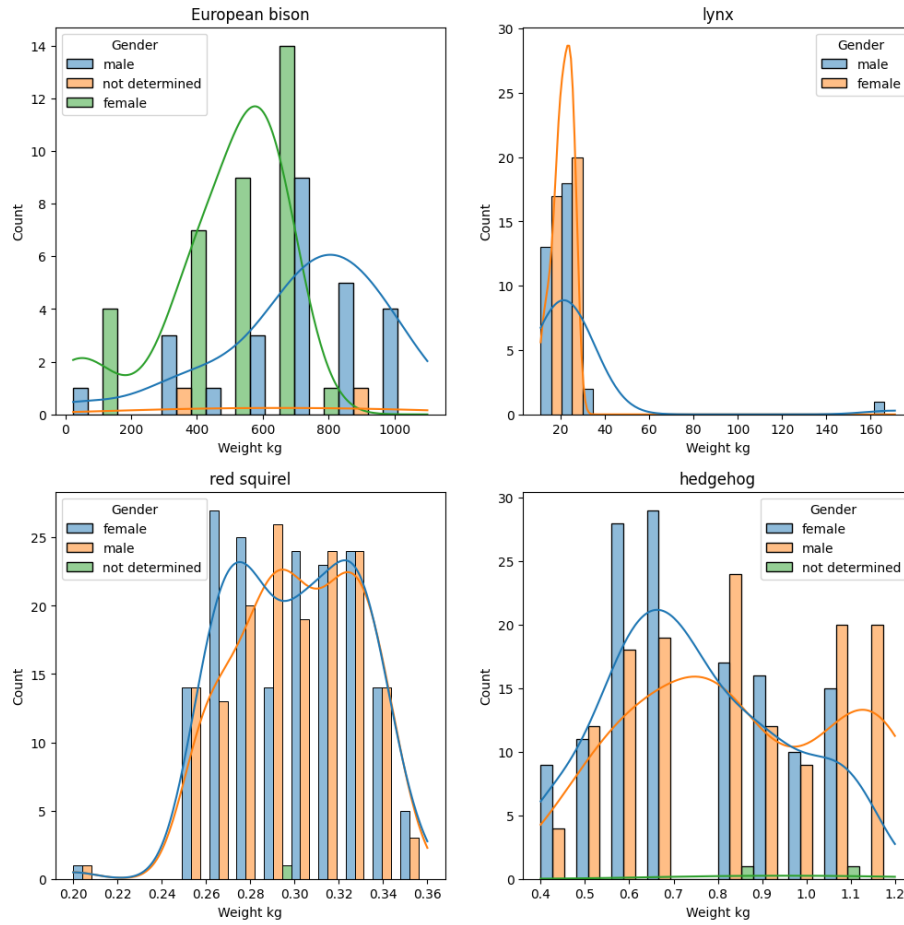


Figure 4: Line plots of observation date by gender and by data contributor

3 Statistical analysis

First of all, I used `pandas.DataFrame.describe()` option to view min, mean, max and quartiles of **wrecked data**.

Right now I greatly sorrow, that I'd thought of choosing [1]. It was a great mistake.

Hence of this, in this section I used **cleaned data** in order of simplicity. It look like this:

References

- [1] Dirty data to clean What's wrong with this dataset.

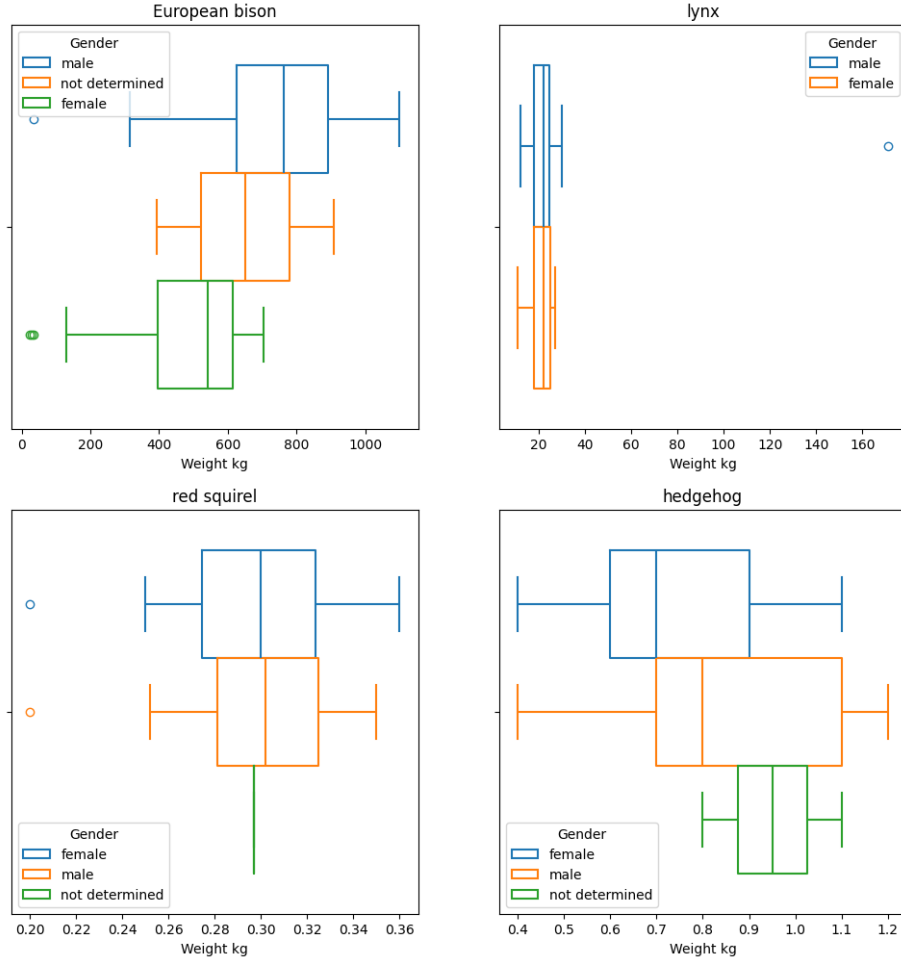


Figure 5: Line plots of observation date by gender and by data contributor

Statistic	Weight (kg)	Body Length (cm)	Animal code	Latitude	Longitude
count	984.000000	984.000000	0.0	913.000000	913.000000
mean	39.745503	39.107724	NaN	49.393369	18.203280
std	156.290076	58.628601	NaN	7.168900	3.899601
min	-0.252000	-19.000000	NaN	-78.582973	11.074008
25%	0.293000	19.000000	NaN	48.186913	14.384559
50%	0.331500	21.000000	NaN	49.560723	18.944015
75%	0.800000	23.000000	NaN	52.212433	21.033243
max	1100.000000	350.000000	NaN	52.853843	34.896734

Table 3: Descriptive statistics for **wrecked data**

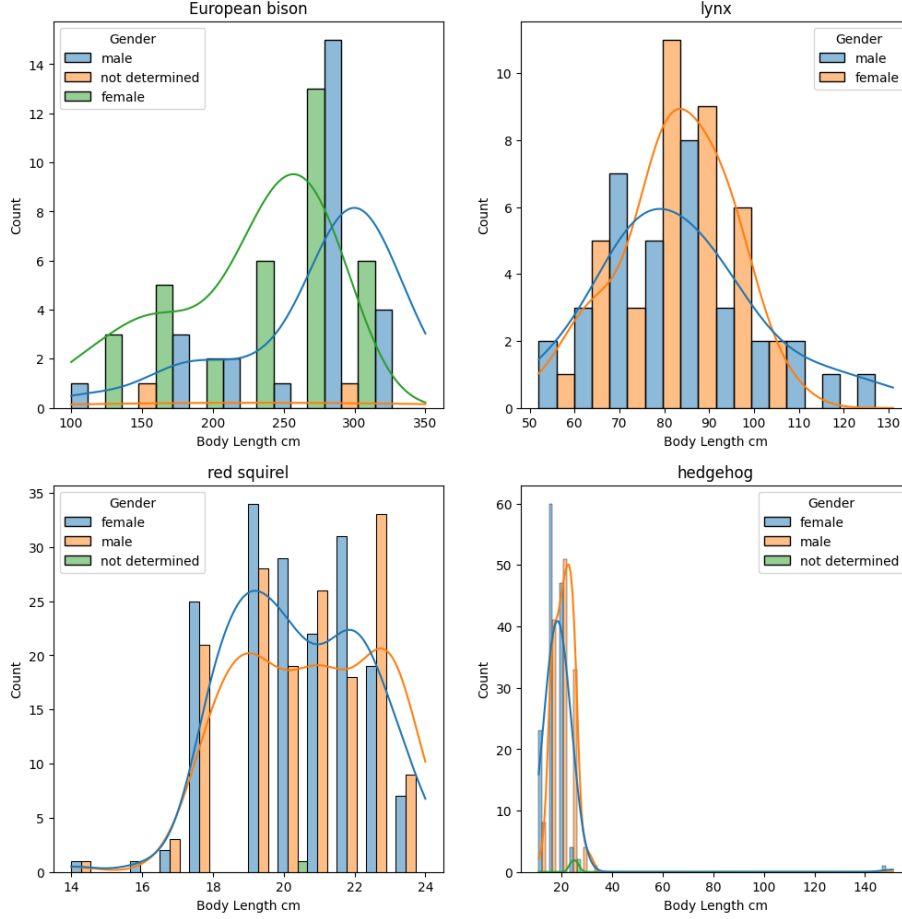


Figure 6: Line plots of observation date by gender and by data contributor

Statistic	Weight (kg)	Body length (cm)	Latitude	Longitude
count	833.000000	833.000000	750.000000	750.000000
mean	46.720602	42.501043	49.743041	17.595264
std	168.843218	62.879887	1.851982	3.881169
min	0.200000	11.000000	47.316383	11.074008
25%	0.297367	19.000000	48.186430	14.348471
50%	0.349000	21.000000	49.545885	18.847488
75%	1.000000	23.000000	52.211980	21.031909
max	1100.000000	350.000000	52.853843	23.919668

Table 4: Descriptive statistics for **cleaned data**

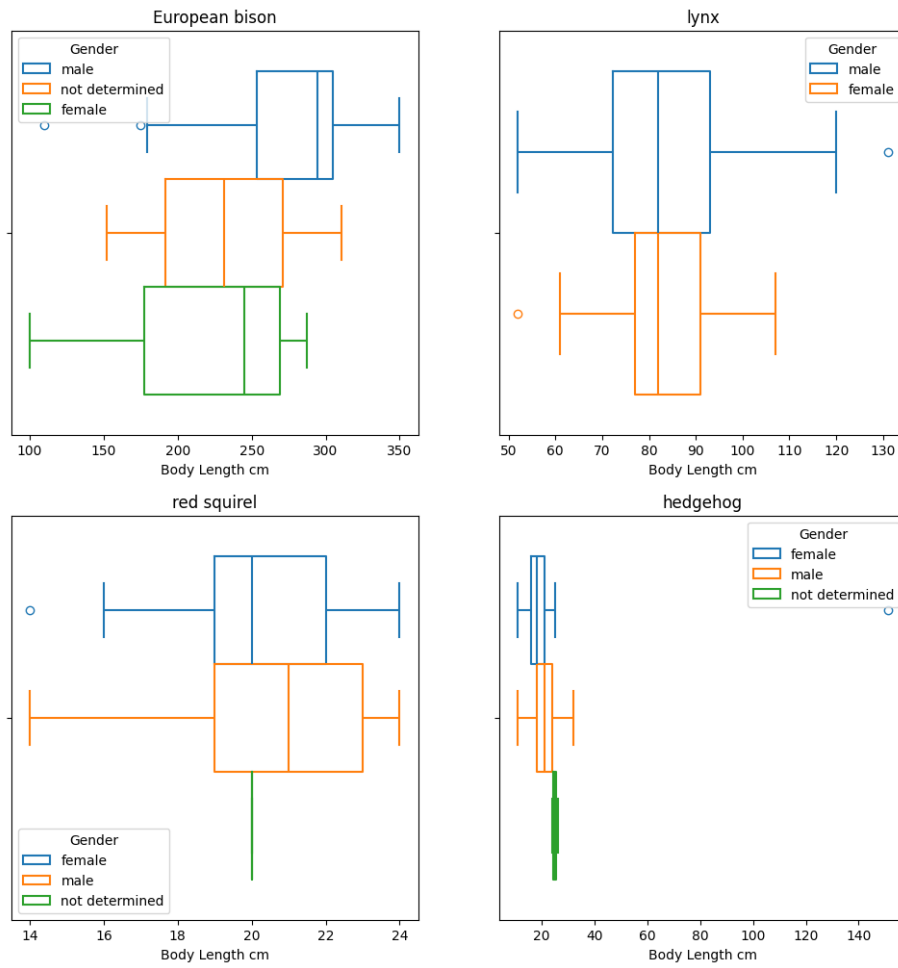


Figure 7: Line plots of observation date by gender and by data contributor